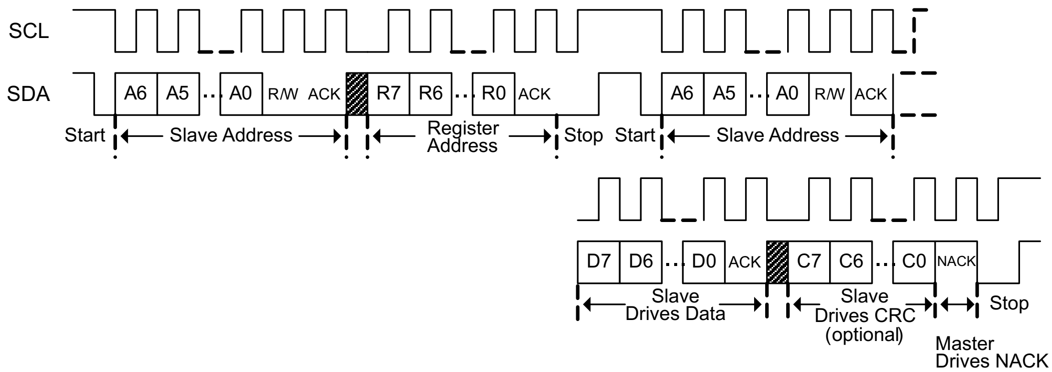

图 8-5. I²C Read Without Repeated Start

8.4 Device Functional Modes

Each BQ769x0 device supports the following modes of operation.

表 8-2. Supported Power Modes

Mode	Description
NORMAL	Fully operational state. Both ADC and CC may be on, or disabled by host microcontroller. OV and UV protection enabled if ADC is on. OCD and SCD enabled. ADC and CC may be disabled to reduce power consumption, and CC may be operated in a "1-SHOT" mode for flexible power savings.
SHIP	Lowest possible power state, intended for pack assembly and/or longterm pack storage. Must see a BOOT signal (> 1 V_{BOOT}) on TS1 pin to boot from SHIP → NORMAL. Note that the device always enters SHIP mode upon POR.

8.4.1 NORMAL Mode

NORMAL mode represents the fully operational mode where all blocks are enabled and the device sees its highest current consumption. In this mode, certain blocks/functions may be disabled to save power—these include the ADC and CC. OV and UV are running continuously as long as the ADC is enabled. The OCD and SCD comparators may not be disabled in this mode.

Transitioning from NORMAL to SHIP mode is also initiated by the host, and requires consecutive writes to two bits in the SYS_CTRL1 register.

8.4.2 SHIP Mode

SHIP mode is the basic and lowest power mode that BQ769x0 supports. SHIP mode is automatically entered during initial pack assembly and after every POR event. When the device is in NORMAL mode, it may enter SHIP by the host controller through a specific sequence of I²C commands.

In SHIP mode, only a minimum of blocks is turned on, including the VSTUP power supply and primal boot detector. Waking from SHIP mode to NORMAL mode requires pulling the TS1 pin greater than V_{BOOT}, which triggers the device boot-up sequence.

To enter SHIP mode from NORMAL mode, the *[SHUT_A]* and *[SHUT_B]* bits in the SYS_CTRL1 register must be written with specific patterns across two consecutive writes:

- Write #1: *[SHUT_A]* = 0, *[SHUT_B]* = 1
- Write #2: *[SHUT_A]* = 1, *[SHUT_B]* = 0

Note that *[SHUT_A]* and *[SHUT_B]* should each be in a 0 state prior to executing the shutdown command above. If this specific sequence is entered into the device, the device transitions into SHIP mode. If any other sequence is written to the *[SHUT_A]* and *[SHUT_B]* bits or if either of the two patterns is not correctly entered, the device will not enter SHIP mode.